Current model (ordinary-matter shell + CoE core) vs. A_F, term by term

worked sheet, 27 Sep 2026

1. Model and where each term comes from. Sedov–Taylor flow $\vec{V} = k\,\vec{x}$ (position from the CoE; Taylor 1950; Sedov 1959), decelerated by a point core M_c at the CoE and a uniform shell Ω_s between $R_{\rm in}$ and $R_{\rm out}$ (fixed in space; Earth and all sources inside it). Spherical symmetry makes the field radial, $\vec{V} = h(q)\,\vec{q}$, with $q = s/D$ (D = CoE–Earth distance). Linear perturbation of the code's integration (from $\tau_i = 0.01$, $Ht \simeq 1$):

$$\frac{h(q)}{k} \simeq 1 - \tfrac{1}{2}\Big[\underbrace{\tfrac{M_c}{M_{\rm crit}(D)}\big(\tfrac{1}{\tau_i}-1\big)q^{-3}}_{\text{core tidal term}} + \Omega_s\Big(\underbrace{\tfrac{1}{2}}_{\text{uniform: } m=0 \text{ only}} \underbrace{-\tfrac{3}{2}r_{\rm in}^2 q^{-2} + r_{\rm in}^3 q^{-3}}_{\text{hole } (r_{\rm in}=R_{\rm in}/D)}\Big)\Big], \qquad \beta \equiv \frac{h'}{h}\Big|_{q=1}, \quad \gamma \equiv \frac{h''}{h}\Big|_{q=1}.$$

Chris's A: $a_f = |\Delta\vec{V}|$/separation with sources at $R = 14.3$ Gpc, $\varepsilon \equiv R/D$, $\mu = \cos b_0 \cos\psi$, $\psi = \ell - \ell_0$ (ℓ_0 = anti-CoE direction). Exactly $a_f \propto |h(q_s)(\hat{x} + \varepsilon\hat{n}) - h(1)\hat{x}|/\varepsilon$; to first order in ε (the higher orders are the r/D terms):

$$a_f^2 \propto 1 + B\mu^2 + \varepsilon\,(P\mu + Q\mu^3), \qquad B = 2\beta + \beta^2, \quad P = \beta(3+\beta), \quad Q = (1+\beta)(\gamma+\beta) - 2\beta.$$

Take the square root, then use $\cos^2\psi = \frac{1}{2}(1+\cos 2\psi)$, $\cos^3\psi = \frac{1}{4}(3\cos\psi + \cos 3\psi)$, $\cos^4\psi = \frac{3}{8} + \frac{1}{2}\cos 2\psi + \frac{1}{8}\cos 4\psi$, $\cos^5\psi = \frac{1}{16}(10\cos\psi + 5\cos 3\psi + \cos 5\psi)$ ($b_0 = 0$ as in the code):

$$\frac{a_f}{\langle a_f\rangle} = 1 + a_1\cos\psi + a_2\cos 2\psi + a_3\cos 3\psi + \ldots, \quad a_2 = \frac{\frac{B}{4} - \frac{B^2}{16}}{m_0}, \quad a_1 = \frac{\varepsilon}{2m_0}\Big[P + \tfrac{3}{4}\big(Q - \tfrac{PB}{2}\big) - \tfrac{5QB}{16}\Big], \quad a_3 = \frac{\varepsilon}{2m_0}\Big[\tfrac{1}{4}\big(Q - \tfrac{PB}{2}\big) - \tfrac{5QB}{32}\Big],$$

with $m_0 = 1 + \frac{B}{4} - \frac{3B^2}{64}$. For a pure point mass ($\gamma = -4\beta$, $\beta \to 3K$) this is the earlier line-of-sight form $K[(3\mu^2-1) + \varepsilon(\frac{9}{2}\mu - \frac{15}{2}\mu^3)]$. The geometry gives only $\cos m\psi$ terms: even harmonics set by β, odd ones by εQ. *Rotation* (the sine factor, $A = a_f[1 - {\rm Amp}\sin(\psi+\varphi)]$) adds $C_1 = a_1 - {\rm Amp}\sin\varphi(1 + \frac{a_2}{2})$, $S_1 = -{\rm Amp}\cos\varphi(1 - \frac{a_2}{2})$, $C_3 = a_3 - \frac{{\rm Amp}\,a_2}{2}\sin\varphi$, $S_3 = -\frac{{\rm Amp}\,a_2}{2}\cos\varphi$ (C_m, S_m: $\cos m\psi, \sin m\psi$ parts). **Latitude:** the code evaluates the model in the Galactic plane only (one direction per degree, no $|b|$ cut, no sector average), so the match is done the same way. A tilt b_0 would give $B \to B\cos^2 b_0$ and ε-terms $\propto \cos b_0, \cos^3 b_0$, degenerate with β and εQ; therefore $b_0 = 0$.

2. Term-by-term solution (Newtonian, series above, point-core shape $\gamma = -4\beta$; exact least-squares in brackets):

Term	A_F coefficient	Analytic expression	Solved value
2ℓ phase	$\ell_2 = 14.96°$	ℓ_0 (axis; $\beta > 0$: fast along it)	$\ell_0 = 14.96°$ (12.8°)
2ℓ amplitude	$C_2 = 0.0734$	$a_2(\beta)$	$\beta = 0.161$ (0.210)
1ℓ (rotation)	$C_1 = 0.0398$, $S_1 = 0.0343$	C_1, S_1 of Sec. 1	Amp $= -0.068$, $\varphi = 1.02$ (-0.072, 1.03)
3ℓ cos	$C_3 = -0.0292$	$a_3(\varepsilon Q) - \frac{{\rm Amp}\,a_2}{2}\sin\varphi$	$\varepsilon Q = -0.308 \Rightarrow \varepsilon = 0.348$, $D = 41$ Gpc (47)
3ℓ sin	$S_3 = 0.0130$	$-\frac{{\rm Amp}\,a_2}{2}\cos\varphi$ (check)	predicted 0.0013: **not produced**
$m = 0$	$A_0 = 0.9965$	$N[1 - \frac{1}{2}{\rm Amp}\,a_1\sin\varphi]$	$N = 0.9971$
mass	–	$\beta \simeq \frac{3K}{1-K}$, $K = \frac{1}{2}\frac{M_c}{M_{\rm crit}(D)}(\frac{1}{\tau_i} - 1)$	$M_c/M_{\rm crit}(D) \approx 0.0010$ (linear; ≈0.0017 nonlinear)

3. Input parameters implied by the term-by-term solution vs. Table 1 of universe_bestfit_thinshell.pdf. Each column keeps that fit's shell assumption. Newtonian Ω_s free: $\Omega_s = 1.012$, thin shell $R_{\rm in} = D - R$, $R_{\rm out} = D + R$. Newtonian Ω_s=0.049: $R_{\rm in} = D - 1.028\,R$, $R_{\rm out} = D + 1.475\,R$. Relativistic: $R_{\rm in} = D - 1.040\,R$, $R_{\rm out} = D + 1.171\,R$. The unknowns are solved from the Sec. 2 equations (2ℓ phase $\to \ell_0$, 2ℓ amplitude $\to \beta \to M_c/M_{\rm crit}$, $1\ell \to$ Amp, φ, 3ℓ cos $\to \varepsilon Q \to D$; C_m, S_m, m = 1–3, by least squares). The $a_m(\beta,\gamma)$ come from the code's exact $h(q)$, not the linear formula. Masses use $H_0 = 70$, $\rho_s = \Omega_s\rho_c$.

	Newtonian, Ω_s free		Newtonian, Ω_s=0.049		Relativistic (LTB)
Quantity	term-by-term	Table 1	term-by-term	Table 1	Table 1
D/R	6.474	6.554	2.589	3.439	2.303
D	2.857×10^{24} km (92.6 Gpc)	2.892×10^{24} km (93.7 Gpc)	1.143×10^{24} km (37.0 Gpc)	1.517×10^{24} km (49.2 Gpc)	1.016×10^{24} km (32.9 Gpc)
$R_{\rm in}$	78.3 Gpc	79.4 Gpc	22.3 Gpc	34.5 Gpc	18.1 Gpc
$R_{\rm out}$	106.9 Gpc	108.0 Gpc	58.1 Gpc	70.3 Gpc	49.7 Gpc
M_c	2.38×10^{54} kg	2.47×10^{54} kg	9.41×10^{52} kg	2.31×10^{53} kg	3.38×10^{54} kg
	$1.20\times10^{24}\ M_\odot$	$1.24\times10^{24}\ M_\odot$	$4.73\times10^{22}\ M_\odot$	$1.16\times10^{23}\ M_\odot$	$1.70\times10^{24}\ M_\odot$
$M_c/M_{\rm crit}(D)$	0.00265	0.00265	0.00164	0.00171	0.0835
$M(<D)$, total	3.62×10^{56} kg	3.72×10^{56} kg	2.29×10^{54} kg	4.56×10^{54} kg	5.04×10^{54} kg
$M_{\rm shell}$, total	8.50×10^{56} kg	8.71×10^{56} kg	1.03×10^{55} kg	1.70×10^{55} kg	6.48×10^{54} kg
M within R	3.35×10^{54} kg	3.35×10^{54} kg	1.62×10^{53} kg	1.62×10^{53} kg	1.62×10^{53} kg
Ω_s	1.012[a]	1.012	0.049[a]	0.049	0.049
ℓ_0	13.2°	12.8°	12.8°	13.0°	13.5°
Amp	−0.1246	−0.1240	−0.0637	−0.0622	−0.0558
φ	1.267 rad	1.264 rad	0.941 rad	0.929 rad	0.848 rad
R^2 vs Migkas	0.909	0.909	0.881	0.893	0.885
R^2 vs A_F	0.987	0.987	0.970	0.977	0.963

[a] Held fixed (that column's shell assumption). The harmonic solve weights C_m, S_m equally, not the 71 points, so its R^2 can sit slightly below the point fit.

4. Model vs. generally accepted values. Model columns use the Table 1 fits. $\langle H\rangle$ is the model's sky-mean expansion rate (mean of $a_f = |\Delta\vec{V}|$/separation over the code's directions at R, with the fitted sine). The fits fix only its shape, so its absolute value is solved from

Migkas et al.'s own H_0 values by fitting $H(\ell) = \langle H\rangle\, A(\ell)/\langle A\rangle$ to them (anchor: their all-combined joint result[a]). All rate-dependent rows follow from $\langle H\rangle$: rates, k, $V_O \propto \langle H\rangle$; masses, densities $\propto \langle H\rangle^2$; $t_0 \propto 1/\langle H\rangle$. Lengths use $R = 14.3$ Gpc.

Quantity	Accepted value [source]	Newt. Ω_s free (0.909)	Newt. Ω_s=0.049 (0.893)	Relativistic LTB (0.885)
H_0 [km s^{-1} Mpc^{-1}]	67.36 ± 0.54 (CMB) [1]; 73.04 ± 1.04 (local) [2]	$\langle H\rangle = 71.8\pm0.5$[a] ($= 0.971\,k$)	$\langle H\rangle = 71.6\pm0.5$[a] ($= 1.012\,k$)	$\langle H\rangle = 71.6\pm0.5$[a,d] ($= 0.956/t_0$)
k (LTB: $1/t_0$) [km s^{-1} Mpc^{-1}]	–	74.0	70.8	74.9
Age / k^{-1}	13.797 ± 0.023 Gyr [1]	$k^{-1} = t_0 = 13.22$ Gyr	$k^{-1} = t_0 = 13.81$ Gyr	$t_0 = 13.05$ Gyr
Ordinary-matter mass within R	1.51×10^{53} kg [1][b]	3.67×10^{54} kg	1.64×10^{53} kg	1.60×10^{53} kg
Total matter mass within R	9.67×10^{53} kg [1][b]	same (no dark matter)	same (no dark matter)	same (no dark matter)
Ordinary-matter density ρ_b	4.20×10^{-28} kg m^{-3} [1]	1.02×10^{-26} kg m^{-3}	4.56×10^{-28} kg m^{-3}	4.46×10^{-28} kg m^{-3}
Ω_b (= Ω_m in the model)	$\Omega_b = 0.0493$, $\Omega_m = 0.3153 \pm 0.0073$ [1]	1.012 (1.053 vs $\langle H\rangle$)	0.049 (0.047 vs $\langle H\rangle$)	0.049 (0.046 vs $\langle H\rangle$)
Critical density ρ_c at $\langle H\rangle$	8.52×10^{-27} kg m^{-3} (H_0=67.36) [1]	9.68×10^{-27} kg m^{-3}	9.63×10^{-27} kg m^{-3}	9.64×10^{-27} kg m^{-3}
Earth (Sun) speed vs. CMB	369.82 ± 0.11 km s^{-1} [3]	$V_O = 6.23\times10^6$ km s^{-1} = $20.8\,c$	$V_O = 3.26\times10^6$ km s^{-1} = $10.9\,c$	$V_O = 2.29\times10^6$ km s^{-1} = $7.6\,c$
ratio V_O/(CMB dipole speed)	1	16,848	8,804	6,199
Local Group speed vs. CMB	620 ± 15 km s^{-1} [3]	– (not modelled)	– (not modelled)	– (not modelled)
Bulk flow within $150\,h^{-1}$ Mpc	395 ± 29 km s^{-1} [4]	6.0 km s^{-1}[c]	0.33 km s^{-1}[c]	not computed
Core mass M_c (vs. accepted M_b in R)	–	2.71×10^{54} kg (18×)	2.33×10^{53} kg (1.5×)	3.34×10^{54} kg (22×)
Mass inside D (same comparison)	–	4.07×10^{56} kg (2,692×)	4.61×10^{54} kg (30×)	4.98×10^{54} kg (33×)

[a] Weighted least-squares fit of $\langle H\rangle A(\ell)/\langle A\rangle$ to Migkas et al. (2021, A&A, 649, A151) all-combined H_0: 66.5 ± 1.0 toward $\ell = 273°$ and 72.8 ± 0.6 for the rest of the sky (model value at the cone centre, and the model mean outside a 75° cone). Across all 11 Migkas 2020/2021 choices (each relation, sample and joint result) $\langle H\rangle$ = 70.4±2.4 / 70.3±2.1 / 70.4±2.0 (mean±s.d., the three fits). Migkas's absolute H_0 values rest on their fiducial calibration ($H_0 = 70$ assumed to calibrate the scaling relations), so this anchor inherits it. [b] Derived from [1]: $\Omega_b h^2 = 0.02237$, $\Omega_m = 0.3153$, $H_0 = 67.36$, times the volume of the $R = 14.3$ Gpc sphere; ratios in brackets are model / this accepted ordinary-matter mass. [c] Mean of $\Delta\vec{V}$ over a $150\,h^{-1}$ Mpc ball around Earth, from the code's exact expansion-rate profile (second order in radius/D); the sine factor is not a velocity. [d] LTB: mean of the local expansion rate over the in-plane directions, $(H_\perp + H_r)/2$; the fit's $z = 0.3$ curve is normalised and carries no absolute rate. LTB values rescaled from the fit (which set the 3D-mean local rate to 70) with the same power laws (approximate in GR; the fitted valley shows the shape does not depend on HD/c).

Sources. [1] Planck Collaboration 2020, A&A, 641, A6 (Planck 2018 VI, TT,TE,EE+lowE+lensing). [2] Riess, A. G., et al. 2022, ApJL, 934, L7 (SH0ES). [3] Planck Collaboration 2020, A&A, 641, A1 (Planck 2018 I; Local Group value after Kogut et al. 1993, ApJ, 419, 1). [4] Watkins, R., et al. 2023, MNRAS, 524, 1885 (CosmicFlows-4). Sedov–Taylor: Taylor, G. I. 1950, Proc. R. Soc. Lond. A, 201, 159; Sedov, L. I. 1959, Similarity and Dimensional Methods in Mechanics (Academic Press).